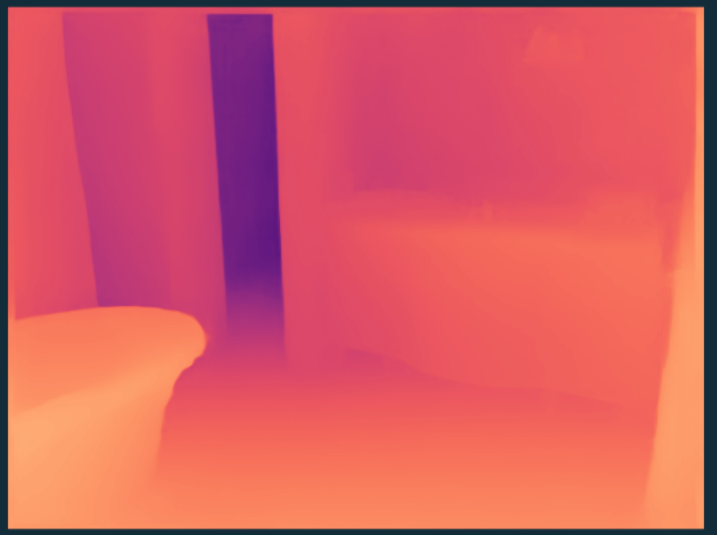
SICHUAN UNIVERSITY

Design and Implementation of a

Color-Constancy-Assisted Monocular Depth Estimation System



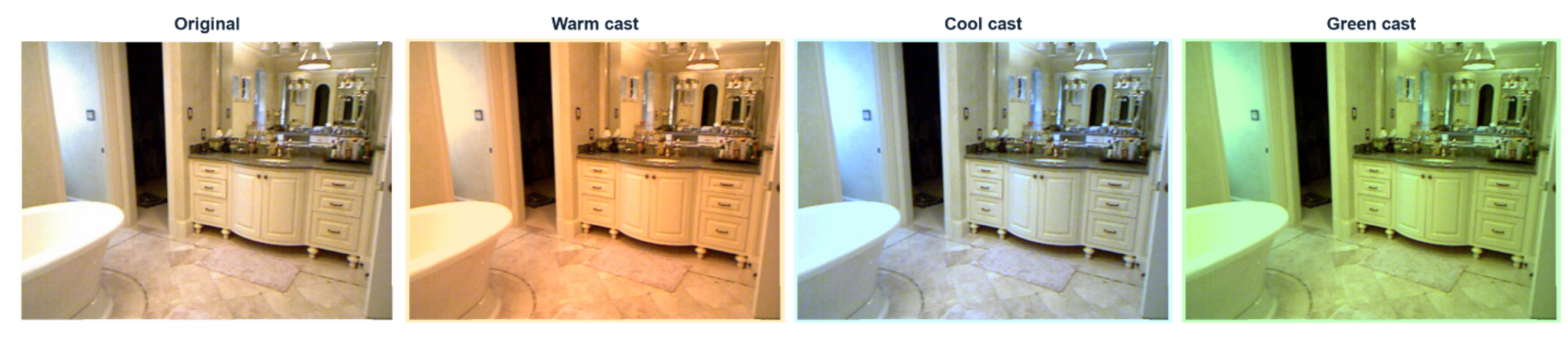
First controlled experiment

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Research question

Can color correction improve depth accuracy when illumination color changes?



The scene and reference depth stay fixed. Only the input color changes.

Implemented experiment

One frozen depth model, paired image comparisons

01

Real RGB image

Aligned reference depth from NYU Depth V2

02

Color processing

7 image conditions
4 correction choices

03

Depth inference

Depth Anything V2
Metric Hypersim Small

04

Accuracy evaluation

Same crop and mask
No scale fitting

1120 saved predictions

8 development images and 32 evaluation images

Dataset and evaluation protocol

Setting	Pilot configuration
Data	NYU Depth V2 RGB and aligned reference depth
Evaluation subset	32 scenes, one image each; 8 separate development scenes
Network input	518 pixel inference setting, aspect ratio preserved
Depth evaluation	Valid reference depth 0.1-10 m, predictions clipped to this range
Image crop	Rows 45:471, columns 41:601 at 480 x 640
Aggregation	Mean of per-image scores, six casts equally weighted

Fixed methods and gains. No parameter fitting to evaluation reference depth.

Color correction baselines

All transformations operate in linear RGB

Gray-World

Channel means

Estimates the illuminant from each channel mean. Applies inverse gains.

White-Patch

Channel maxima

Estimates the illuminant from each channel maximum. Applies inverse gains.

Gray-World half

Square-root gains

Uses half the correction strength in log-gain space. Strength fixed at 0.5.

Every gain transform preserves mean luminance before clipping.

Clean-input accuracy

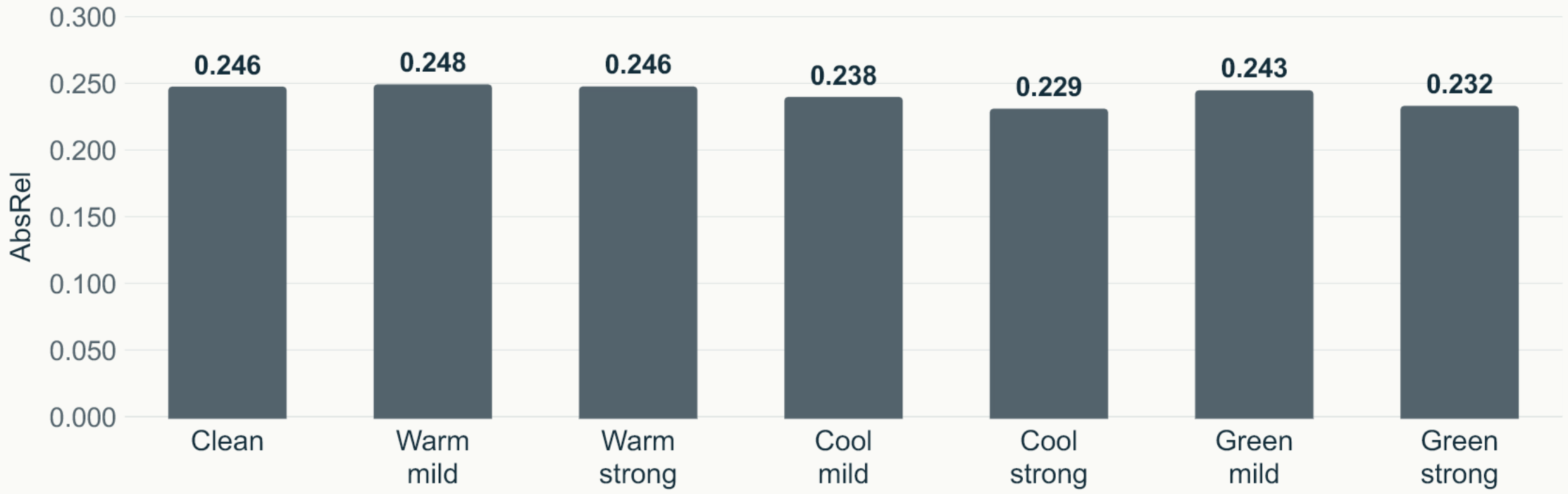
32 evaluation images, without an added color cast

Processing	AbsRel	RMSE (m)	delta1 accuracy
No correction	**0.246**	**0.796**	**63.1%**
Gray-World	0.242	0.781	64.0%
White-Patch	0.246	0.796	63.1%
Gray-World half	0.244	0.789	63.5%

AbsRel and RMSE: lower is better
delta1: fraction of pixels within a depth ratio of 1.25, higher is better

Sensitivity to color casts

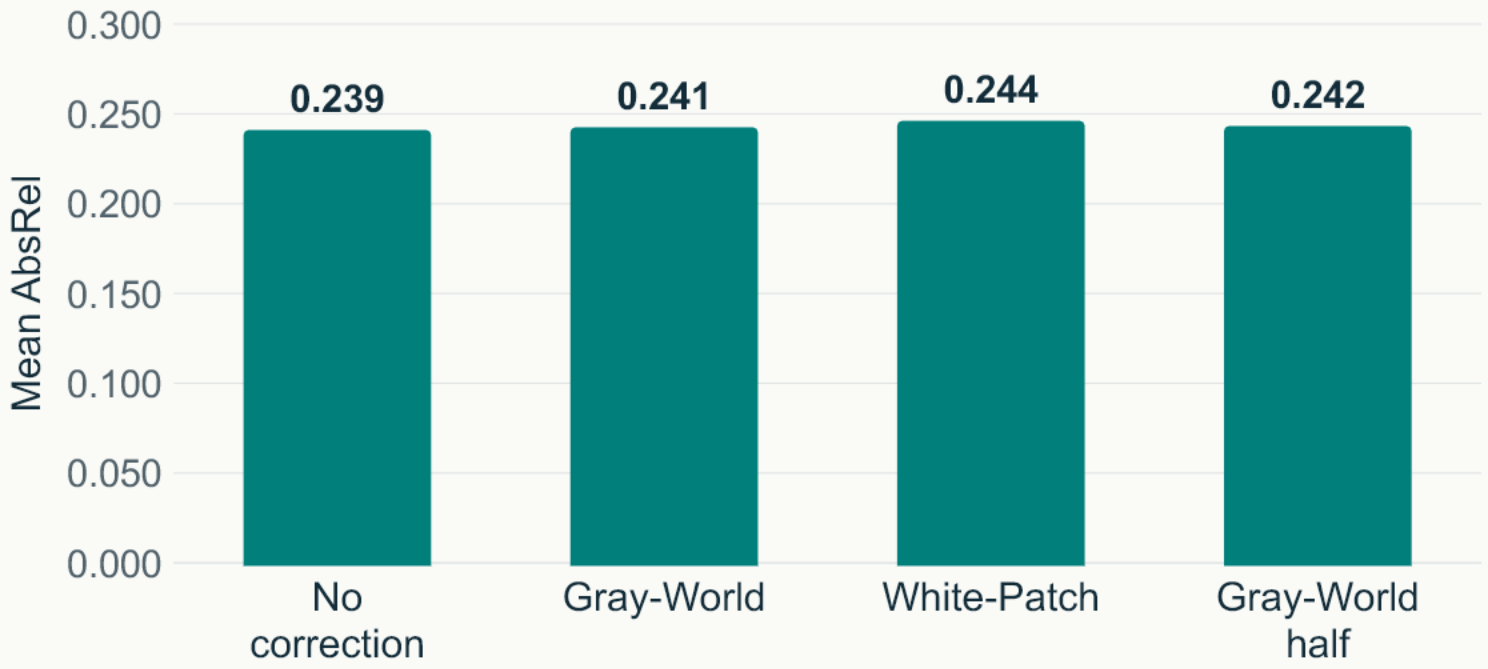
No correction, mean per-image AbsRel across 32 images



Mild and strong casts use fixed channel gains with the same reference depth.

Correction performance across color casts

Fixed baselines, six synthetic casts, equal weight for each cast and image



Lowest among correction baselines

Gray-World

0.6%

mean error increase against no correction

19/32 images improve when averaging their six casts

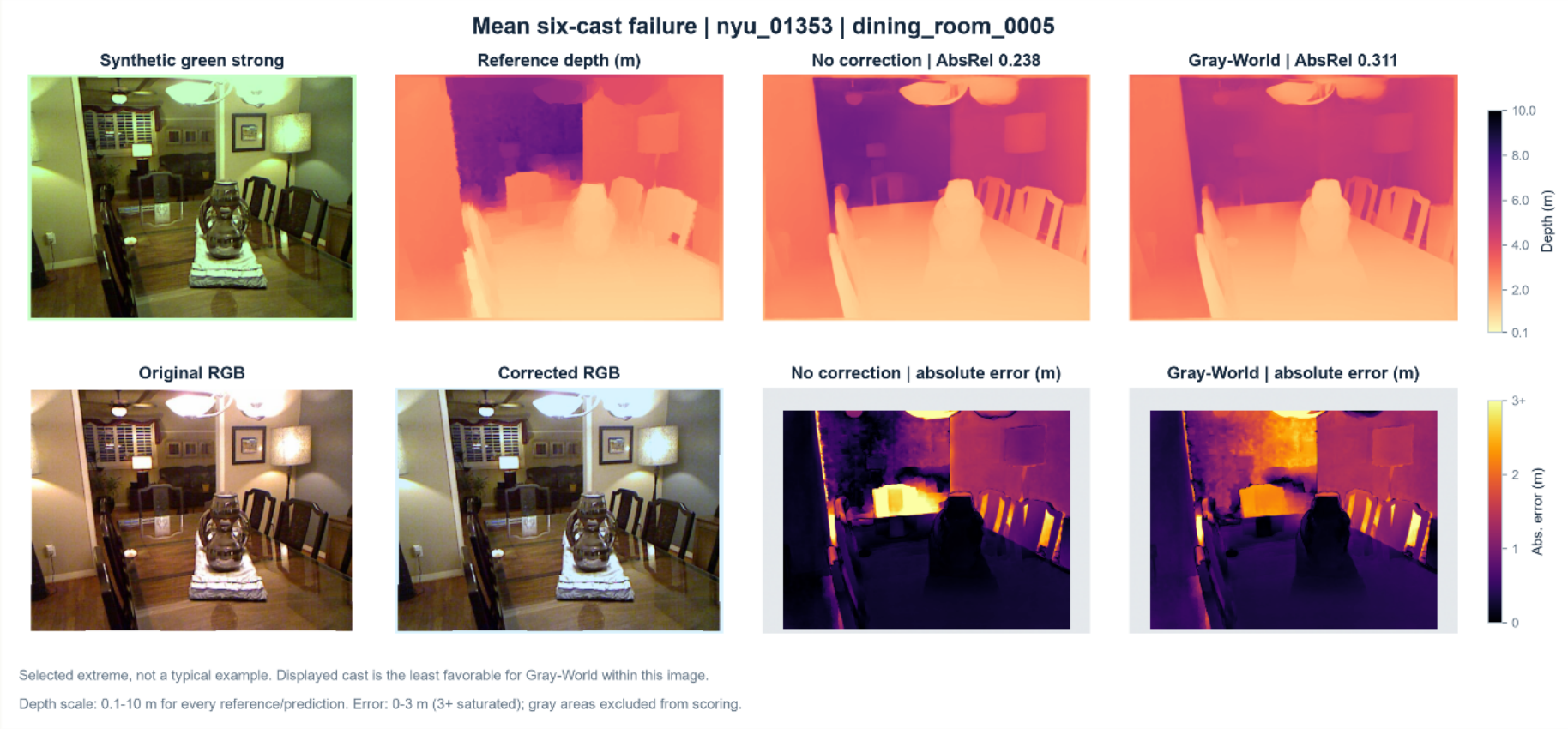
Paired AbsRel reduction -0.0015, 95% bootstrap CI [-0.0061, 0.0025]
The interval crosses zero. This pilot does not establish an average benefit.

Visual comparison of saved depth predictions



The reference and predicted depth maps share the same color scale.

Failure analysis



A global correction can mistake scene color for illumination.

Laptop execution and reproducibility

Item	Completed run
Device	Intel Core Ultra 7 255H, CPU inference
Model	Depth Anything V2 Metric Hypersim Small, 24.8M parameters
Compute setting	8 CPU threads, input setting 518
Predictions	1120 numerical depth arrays saved
Mean inference time	1.13 seconds per image condition
Traceable outputs	Manifest, metrics CSV, checkpoint hash, figures and source code

The project runs in its own Python environment and opens as a VS Code workspace.

Findings and next experiments

Observed in this pilot

The complete paired evaluation runs locally.

Gray-World has 0.6% higher mean cast error.

Its mean clean-input error is 1.7% lower.

Next experiments

Expand the fixed evaluation subset and test a second depth model.

Separate clipping effects from color sensitivity.

Test real lighting changes before designing an adaptive correction rule.

Initial pretrained-model pilot. A novel method and broader validation remain future work.